INTERNATIONAL MONETARY FUND

REPUBLIC OF UZBEKISTAN

April 6, 2021

STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

Approved By
Thanos Arvanitis and Uma Ramakrishnan (IMF) and Marcello Estevão (IDA)

Prepared by the International Monetary Fund and the International Development Association

Uzbekistan: Debt Sustainability Analysis	
Risk of External Debt Distress	Low
Overall Risk of Debt Distress	Low
Granularity in the Risk Rating	Not Applicable
Application of Judgement	No

Under the baseline scenario, the path of public and publicly guaranteed (PPG) external debt is expected to remain around 40 percent of GDP over the medium term.[1] This is about 7 percentage points of GDP higher than under the previous DSA of May 2020. Due to the COVID-19 shock, the government's overall fiscal deficit and disbursements of guaranteed debt are expected to be higher and GDP lower over the next five years. Uzbekistan's total and PPG external debt are projected to peak at 64 and 43 percent of GDP, respectively, in 2022 before declining modestly out to 2026 and then stabilizing.

Staff assesses that Uzbekistan remains at a low risk of external debt distress.[2] Under the baseline scenario, PPG external debt indicators remain well below relevant thresholds. Under the most extreme scenario (worse-than-expected exports), the debt service-to-exports ratio would breach its threshold. This is discounted as it occurs in only one year (2024) due to repayment of a Eurobond, then the ratio returns and remains below the threshold thereafter. Shock scenarios with higher primary deficits or higher contingent liabilities arising from state enterprise debt would raise risk indicators modestly. The probability of these risks being realized is about the same as in the May 2020 DSA.

Under the authorities' plans, which include an annual limit on PPG debt commitments and a limit on the overall PPG debt stock, the PPG debt-to-GDP ratio will remain around 40 percent of GDP over the medium term. Strong foreign exchange reserves and low rollover risk (due to the long-term maturity of debt), mitigate the potential risk of debt distress. To limit risks to its strong external position, the government should continue to carefully manage public and external borrowing, improve public investment management and coordination, and be cautious on granting government debt guarantees.

---

[1] This DSA was prepared jointly by IMF and World Bank staff and is based on the Joint Bank-Fund Low-Income Country Debt Sustainability Analysis (LIC-DSA) methodology.

[2] Uzbekistan's Composite Indicator score is 3.16 based on the October 2020 WEO and 2019 CPIA indicating a strong debt carrying capacity (see Text Table 4).

BACKGROUND

A. Public Debt Coverage

1. Public debt coverage is broad (see text table 1). Coverage includes debt of the central government, state and local governments, the pension fund, extra-budgetary funds, and debt of state-owned enterprises (SOEs) guaranteed by the government. Public debt does not include non-guaranteed debt of SOEs and joint ventures (about 6 percent of GDP in 2020 down from 7 percent in 2019). The authorities are improving coverage and have made steady progress over the last few years with technical support from the IMF and World Bank. The government established a Debt Management Office (DMO) in the Ministry of Finance and has been implementing Debt Management Financial Analysis System (DMFAS) software from the United Nations Conference on Trade and Development (UNCTAD). Private external debt is primarily comprised of borrowing by state owned banks and by joint ventures of SOEs.

Text Table 1. Debt Coverage

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

2. Contingency stress tests are based on standard parameters (see Text Table 2). For the financial market shock, staff used the default value of 5 percent of GDP. But for a shock to SOE debt, staff raised the value from 2 to 7 percent of GDP. This reflects the large share of SOEs in economic activity and the potential for contingent liabilities to be realized in the future due to the rapid rise in credit growth over the last few years. Currently, liabilities arising from Public Private Partnership (PPP) projects are not large, although the government has put in place a legal framework and plans to scale up PPPs significantly over the medium-term.

Text Table 2. Magnitude of Contingent Liability Shock

1	The country's coverage of public debt	The general government, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	7.0	
4	PPP	35 percent of PPP stock	0.0	No PPPs currently
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		12.0	

B. Background on Debt and Short-Term Developments

3. Uzbekistan's public and external debt levels have risen rapidly in recent years from low levels in the past. Total PPG debt rose from US$ 9.9 billion (20.3 percent of GDP) at end-2018 to US$ 21.0 billion (37.8 percent of GDP) at end-2020. Similarly, total external debt (the sum of public and private external debt) rose from US$ 17.0 billion (34.3 percent of GDP) at end-2018 to US$ 32.9 billion (58.4 percent of GDP) at end-2020. These increases were primarily due to higher PPG external debt, which makes up almost two-thirds of external debt and 98 percent of total PPG debt.

- In 2019, public external borrowing was driven by project financing. PPG external debt disbursements were US$ 6.3 billion (11.0 percent of GDP). US$ 4.4 billion (about 8 percent of GDP and 70 percent of total disbursements) financed projects and US$ 1.8 billion (about 3 percent of GDP and 30 percent of total disbursements) was used as budget support. This included proceeds from Uzbekistan's first issuance of Eurobonds, worth US$ 1 billion. Recognizing that this level of disbursement would be difficult to sustain, the government instituted an annual limit on PPG external debt commitments of US$ 4 billion.
- In 2020, to address the COVID crisis, the government increased expenditures for healthcare, social assistance, investment, and other support to the economy. The annual limit on PPG external debt commitments was eased, and disbursements were US$ 5.3 billion. The overall fiscal deficit increased modestly, from 4 percent of GDP in 2019 to 4½ percent of GDP in 2020. The share of external borrowing used as budget support rose to about 50 percent. In late 2020, the government successfully placed about US$ 750 million in sovereign bonds. This included a US$555 million, 10-year Eurobond and an UZS 2 trillion (US$ 192 million), 3-year som denominated bond. The US dollar Eurobond was issued at a 3.7 percent interest rate. In early February 2021, it was trading with a yield of 3.6 percent, about 240 basis points above US Treasuries. The som denominated bond was issued at 14.5 percent interest rate and was trading at about 14 percent in early February 2021.

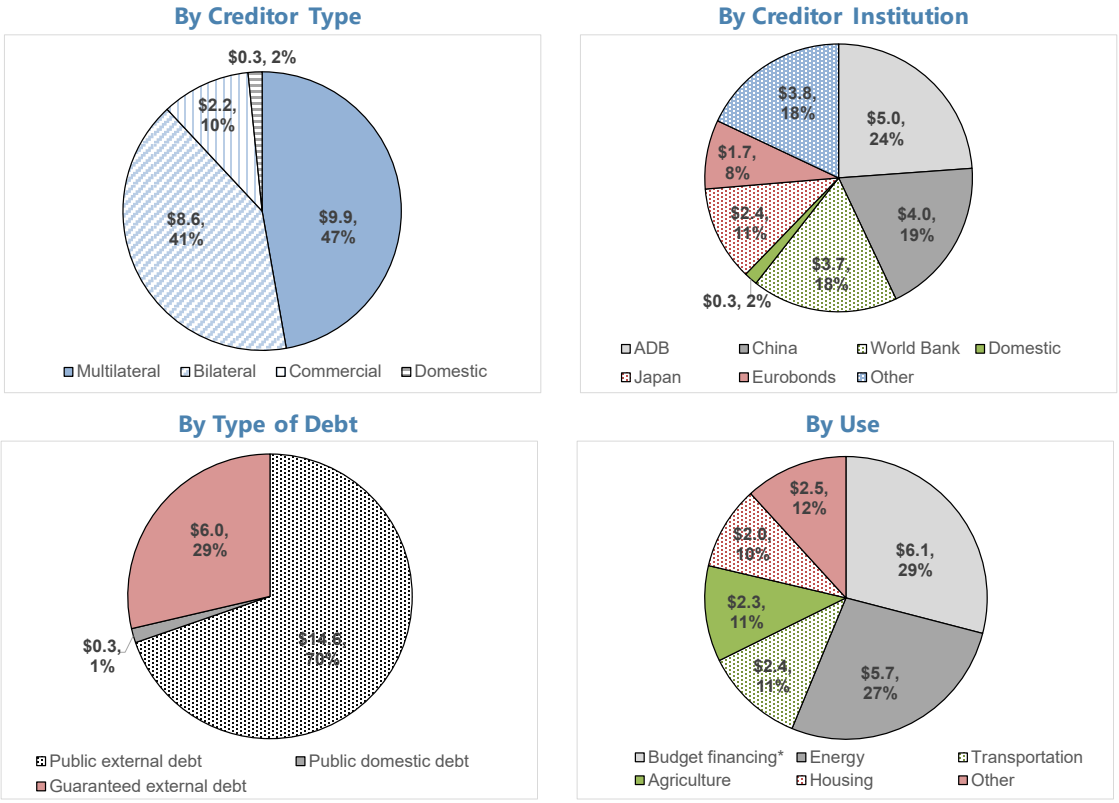
4. Most of Uzbekistan's public debt has been provided by multilateral and official bilateral creditors and has been used to fund infrastructure projects and government deficits (see Text Figure 1).

- Multilateral and bilateral creditors have each provided about 40 percent of Uzbekistan's PPG debt financing. Among multilateral institutions, the Asian Development Bank and World Bank are the largest creditors. Among bilateral donors, China and Japan are the largest. Two thirds of PPG debt is comprised of public debt, while one third is publicly guaranteed debt.
- In terms of use, about a quarter of external financing has been used as general budget support. The remainder has financed infrastructure projects. With substantial reserves of natural gas and oil, the energy sector has received about a quarter. But significant financing has also gone to projects in the agriculture, housing, and transportation sectors.
- Domestic public debt is comprised of government bills and bonds. As of end-2020, domestic debt securities amounted to US$ 340 million (0.6 percent of GDP). Seventy percent was comprised of short-term bills with maturities of one year or less. At end-2020, outstanding

government bonds had original maturities ranging from 1½ to 3 years, with average maturity of 2.2 years.

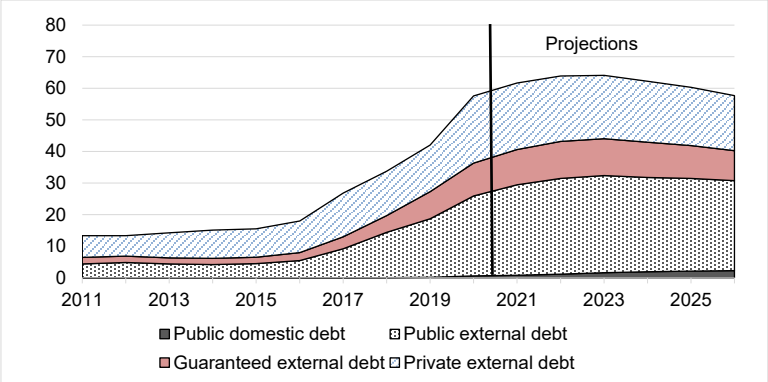
Text Figure 1. Composition of Public and Publicly Guaranteed Debt, end-2020

Total = US$ 21.0 Billion (37.8% of GDP)
External = US$ 20.6 billion (37.2% of GDP)
Domestic = US$ 0.3 billion (0.6% of GDP)



* includes external donor support and domestic securities.

Historical and Projected Evolution of Debt
(percent of GDP)



Source: Debt Management Office of the Ministry of Finance and IMF staff calculations.

5. Private external debt has also risen significantly over the last few years. With trade and foreign exchange liberalization in 2017, many firms increased purchases of capital goods, which jumped by US$ 3 billion in 2018 and US$ 2 billion in 2019, some of which was financed by state banks that borrowed abroad. In 2020, private external debt rose by US$3.7 billion (6½ percent of GDP) and stood at US$ 12.2 billion (21 percent of GDP) at end-2020. State banks significantly expanded external borrowing in 2020 to finance domestic credit expansion as the government reduced policy lending.

UNDERLYING ASSUMPTIONS AND COUNTRY CLASSIFICATION

A. Assumptions for the Macroeconomic Forecast

6. Compared to the May 2020 DSA, macroeconomic conditions have moderately worsened due to a more prolonged COVID crisis. The weaker outlook is expected to persist in 2021. In particular:

- **Growth and Inflation**: Real GDP growth in 2020 was 1.6 percent, about the same as forecast in May 2020. Projections for 2021 and 2022 are now about 1½ and half percent lower, respectively, than forecast in May 2020 as the COVID pandemic is expected to affect growth further into 2021 and 2022. For 2021-2026, the average growth projection remains about 5.5 percent, then declines to 5 percent for 2027-2041. Consumer price inflation is projected to remain around 10 percent with an increase in public wages in 2021 and increases in administrative energy prices in 2022. Thereafter, central bank policies are projected to bring consumer price inflation close to their medium-term target of 5 percent
- **Fiscal outlook**: The government's overall fiscal deficit was 4.4 percent of GDP in 2020 somewhat smaller than forecast in May 2020. Overall revenues were stronger than expected, due in part to higher gold prices which boosted mining profits. Overall expenditures were about as forecast and included additional spending on healthcare, social protection, investment, and support to businesses during the COVID crisis. Staff projects that the overall fiscal deficit will be higher in 2021 as some spending is carried over from 2020 and over the medium-term as consolidation is pushed back. In particular, the overall fiscal deficit is projected to rise to 5½ percent of GDP in 2021, before returning to the government's target of 2 percent of GDP over the medium-term, as revenues increase modestly and policy lending decreases.
- **External outlook**: The current account in 2020 improved to a deficit of 5.4 percent of GDP, compared to 5.8 percent of GDP in 2019 and staff's May 2020 forecast of a deficit of 9.6 percent of GDP. While exports and remittances fell compared to 2019, this was more than offset by the decline in imports. A decline in natural gas exports was partially offset by a higher price of gold, Uzbekistan's largest export. The current account deficit is projected to widen in 2021 as imports rebound. Over the medium term, the current account deficit is projected to improve to 5 percent of GDP in 2026 due to a recovery in goods and services exports.

- **Financing Strategy**: In the near-term, international creditors are expected to continue to provide the majority of financing. Financing from external bond issuance and the Fund for Reconstruction and Development will remain at about the same level as in 2020. External budget support (primarily from International Financial Institutions) is expected to decline. Going forward, the government plans increase the annual limit on debt commitments to about US$ 5 billion in 2021 and US$ 5.5 billion over the medium term but will include gross domestic debt issuance in the limit. Disbursements are assumed to be comprised of 30 percent multilateral, 25 percent bilateral, 20 percent sovereign, and 25 percent domestic debt. Domestic debt issuance is assumed to be about one-half government bills and one-half bonds, rising to 2¼ percent of GDP by 2026. Over the long-term, as the local bond market develops, the share of financing from domestic securities is assumed to gradually increase to 5½ percent of GDP by 2041.

Text Table 3. Comparison of Key Macroeconomic Assumptions

DSA Vintage:	Actual	May 2019 Art IV		May 2020 RCF/RFI		Current	
Key macroeconomic variables (annual averages)	2015-20	2019-24	2025-39	2020-25	2026-40	2021-26	2027-41
				(percent change)			
Real GDP growth	5.1	5.9	6.0	5.2	5.1	5.4	5.1
GDP deflator (UZS)	2.5	12.4	6.7	8.4	5.5	7.6	3.1
Nominal GDP (UZS)	7.8	19.1	13.1	14.0	10.8	13.4	8.3
Exports of goods & services (USD)	1.2	9.2	7.7	10.6	7.6	14.9	7.7
Fiscal balance				(percent of GDP)			
Revenues & grants	26.4	25.7	26.1	25.3	25.7	27.6	28.6
Primary expenditure	28.5	27.1	27.6	28.2	27.5	30.6	30.7
Primary deficit	2.2	1.4	1.4	2.8	1.8	3.0	2.0
Interest expenditure	0.1	0.2	0.2	0.2	0.3	0.5	0.5
Overall fiscal deficit	2.3	1.6	1.6	3.1	2.1	3.4	2.5
External balance				(percent of GDP)			
Non-interest current account deficit	1.5	4.1	3.4	4.8	7.6	3.7	3.0
Current account deficit	2.4	5.0	4.2	6.4	4.3	5.5	4.8

Source: Authorities' data and IMF and World Bank staff estimates

B. Realism Tools

7. Realism tools suggest Uzbekistan is an outlier relative to peers, reflecting the rapid increase in debt, impact of the COVID crisis, and the authorities' reforms.

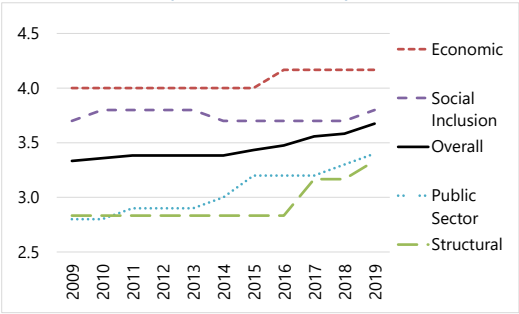
- **Forecast errors**. The growth of Uzbekistan's public and external debt is above the 75th percentile for low income countries (see Figure 3). For PPG external debt, the largest contribution has come from prices and the exchange rate. This is not surprising as the opening of the trade and foreign exchange regimes in 2017 was accompanied by a 50 percent depreciation of the official exchange rate. For overall PPG debt, the largest contributions have come from primary deficits and other debt creating flows (i.e. public external borrowing for projects). In both cases, the increase in debt creating flows is expected to drop from close to 30 percent of GDP over the last 5 years to about 5 percent over the next five years. This reflects a stabilization of the exchange rate and a return to growth following the pandemic.

- **Fiscal adjustment**. The realism tools suggest that the projected fiscal adjustment over the next three years is feasible as Uzbekistan's adjustment is not far from the mean for adjustments in other countries (see Figure 4). While history suggests fiscal consolidation could lower growth, previous experience may provide less guidance given the special features of the COVID shock. In particular, as the crisis subsides global and domestic growth are expected to rebound. With greater external demand, the negative impact of fiscal consolidation on growth is likely to be less than in more normal times.
- **Investment and growth.** For 2020, government investment was modestly higher and private investment modestly lower than projected in May 2020. This reflects additional government stimulus spending and the poorer outlook for private investment as the pandemic continued longer than previously expected. Over the medium-term, staff's current projections for government and private investment are similar to those in May. Government investment is expected to level off as a share of GDP, while economic reforms provide incentives for private firms to raise investment.

C. Country Classification and Scenario Stress Tests

8. Uzbekistan's debt-carrying capacity is assessed as strong (see Text Table 4). Uzbekistan's Composite Indicator (CI) score is 3.16.[3] The strong CI score reflects high average international reserves (about 60 percent of the 10-year average of imports) and high average Country Policy and Institutional Assessment (CPIA) ratings. It represents a modest decline from Uzbekistan's CI score of 3.21 at the time of the May 2020 DSA. The decline was primarily caused by lower projected global growth due to the more prolonged impact of the COVID pandemic. Remittances and scores for domestic components of the CPIA rose. Uzbekistan's CPIA score rose due to improvements in the business environment, revenue mobilization, public sector transparency, and equitable use of public resources. In May 2020, staff projected remittances would fall significantly in 2020 as the pandemic prevented Uzbekistan workers from travelling abroad to work. However, remittances were higher than expected and staff has revised the projections upwards.

Text Figure 2. Country Policy and Institutional Assessment, 2009-2019
(Score out of 6)



Source: World Bank.

[3] The CI is based on the IMF's World Economic Outlook (WEO) of October 2020 and the World Bank's Country Policy and Institutional Assessment (CPIA) for 2019. The DSA-LIC framework uses the CI to capture factors affecting a country's debt carrying capacity. The CI is calculated using a weighted average of a country's CPIA score, real GDP growth, remittances, foreign exchange reserves, and global growth averaged over 5 years of historical data and 5 years of projections.

EXTERNAL DSA

9. Staff assesses Uzbekistan's risk of external debt distress as low. From 2020 to 2022, PPG external debt is projected to rise from 37 to 43 percent of GDP, before gradually declining over the medium-term (see Table 1). Total external debt is driven by PPG external debt (about two-thirds of the total) and is projected to peak at about 64 percent of GDP in 2022. Under the baseline scenario, sustainability indicators stay well below risk thresholds.

10. PPG external debt is most vulnerable to a shock to exports (see Figure 1 and Table 3). Under a one standard deviation shock to exports, the debt service-to-exports ratio would breach its threshold in 2024 as a US$ 500 million Eurobond issued in 2019 comes due. As a temporary breach, it is discounted from the risk ratings. The most significant other stress scenarios are: (i) a combination of shocks and (ii) a decline in remittances and FDI. But compared to an export shock, these scenarios would have much less of an impact and debt indicators would remain below risk thresholds. Under the market financing scenario, debt indicators would rise slightly above the baseline (see Tables 3 and 4) and indicators suggest gross financing need and spreads would remain well below thresholds (see Figure 5).

Text Table 4. Composite Indicator of Debt Carrying Capacity

Debt Carrying Capacity	Strong

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Strong	Strong 3.16	Strong 3.21	Strong 3.17

Applicable thresholds

EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	240
GDP	55
Debt service in % of	
Exports	21
Revenue	23

TOTAL public debt benchmark	
PV of total public debt in percent of GDP	70

OVERALL RISK OF PUBLIC DEBT DISTRESS

11. Staff assesses Uzbekistan's overall risk of public debt distress as low. Overall PPG debt is projected to rise from 38 percent of GDP in 2020 to 44 percent of GDP in 2022, before gradually declining out to 2026 and then stabilizing around 40 percent of GDP (see Table 2). This is line with the government's plan to implement fiscal rules, including a debt to GDP limit of 60 percent of GDP. While the overall PPG debt ratio is projected to remain stable, the composition would shift, with domestic securities gradually rising to about 5½ percent of GDP in 2041. PPG external debt would correspondingly decline modestly as a share of GDP. Multilateral and official bilateral creditors are expected to provide the majority of financing over the medium term, although the share of budget support should decline.

12. Stress tests suggest Uzbekistan's PPG debt ratios are robust to a wide range of shocks (see Figure 2 and Table 4). Even under the most extreme scenario (a decline in exports), the PV of debt-to-GDP ratio remains more than 20 percent of GDP below the public debt benchmark. Greater use of domestic debt would reduce the risk of reliance on foreign currency external debt. The authorities will need to carefully manage the maturity structure and currency composition of public debt to mitigate the burden of debt service. The stress test results in this DSA are similar to those of the May 2020 DSA.

RISK RATING AND VULNERABILITIES

13. Uzbekistan is at low risk of debt distress, but a rapid increase in debt in recent years underlines the need to carefully manage both external and public debt and act if indicators worsen significantly. Uzbekistan needs significant investment to achieve its development goals. These include raising growth, creating high quality jobs, upgrading infrastructure, and improving social support systems. At the same time, a continuation of the rapid increase in debt that has occurred over the last several years could quickly exhaust Uzbekistan's fiscal space and raise vulnerabilities. The authorities have already taken steps to limit a further rapid increase in debt. They have implemented annual limits on new commitments of PPG debt and are drafting a public debt law which will limit overall PPG debt to 60 percent of GDP. Staff's analysis indicates that with the annual limits on PPG debt, debt ratios will peak in 2022 and subsequently decline. Strong FX reserves and low rollover risk, due to the long-term maturity of debt, mitigate the risk of debt distress. To limit risks to its strong external position, the government should continue to carefully manage public and external borrowing, improve public investment management and coordination, and be cautious about granting government debt guarantees. The government is also encouraged to include non-guaranteed debt of state enterprises in the coverage of public debt, as the state enterprise sector comprises a significant share of the economy.

AUTHORITIES' VIEWS

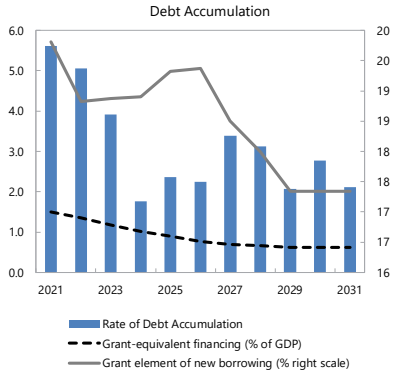
14. The authorities generally agreed with the outlook and emphasized their commitment to maintain sustainable public and external debt. They noted substantial progress in improving debt statistics, including via the DMFAS system. The Ministry of Finance's Debt Management Office has provided better tracking, analysis, and management of domestic and external debt. Nonetheless, they are cognizant of the potential risks arising from the increase in public debt, as well as from contingent liabilities. To address these issues, they noted they had recently undergone an assessment of their public investment management system with a view to improving project selection and monitoring and enhancing intra-governmental coordination on investment and debt management. They emphasized their commitment to ensure debt sustainability and pointed to concrete policy actions, including the implementation of annual limits on PPG debt commitments and the preparation of a debt law that would introduce a cap on PPG debt as a share of GDP.

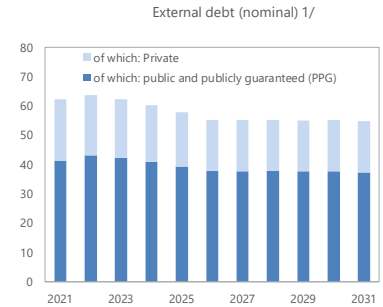
Table 1. Uzbekistan: External Debt Sustainability Framework, Baseline Scenario, 2020-2041

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**58.4**	**62.3**	**63.8**	**62.3**	**60.3**	**57.8**	**55.2**	**54.8**	**52.0**	**26.3**	**57.9**
of which: public and publicly guaranteed (PPG)	**37.2**	**41.3**	**43.1**	**42.3**	**41.0**	**39.4**	**37.8**	**37.3**	**34.5**	**15.0**	**39.4**
Change in external debt	14.5	3.8	1.5	-1.5	-2.0	-2.5	-2.6	-0.5	-0.5		
Identified net debt-creating flows	**3.0**	**1.2**	**0.4**	**-0.7**	**-1.1**	**-1.3**	**-1.5**	**-1.3**	**-1.1**	**-1.6**	**-0.9**
Non-interest current account deficit	**4.1**	**5.1**	**4.1**	**3.5**	**2.8**	**3.3**	**3.4**	**3.0**	**3.0**	**-0.3**	**3.3**
Deficit in balance of goods and services	14.1	15.2	14.6	13.7	13.1	12.6	12.0	12.0	12.0	**7.7**	**12.8**
Exports	24.9	28.1	30.1	31.4	31.8	32.0	32.2	32.2	32.2		
Imports	39.0	43.3	44.8	45.1	44.9	44.5	44.2	44.2	44.2		
Net current transfers (negative = inflow)	-9.0	-9.2	-9.0	-8.6	-8.1	-7.6	-7.1	-7.1	-7.1	**-5.0**	**-7.7**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	-1.0	-0.9	-1.5	-1.6	-2.2	-1.7	-1.5	-1.9	-1.9	**-3.0**	**-1.8**
Net FDI (negative = inflow)	**-2.4**	**-2.5**	**-2.5**	**-3.0**	**-3.2**	**-3.4**	**-3.5**	**-3.5**	**-3.5**	**-1.9**	**-3.2**
Endogenous debt dynamics 2/	**1.4**	**-1.5**	**-1.2**	**-1.2**	**-0.6**	**-1.2**	**-1.5**	**-0.8**	**-0.6**		
Contribution from nominal interest rate	1.4	1.3	1.8	1.9	2.4	1.8	1.4	1.8	1.8		
Contribution from real GDP growth	-0.7	-2.7	-3.0	-3.2	-3.1	-3.0	-2.9	-2.6	-2.5		
Contribution from price and exchange rate changes	0.7	...	...	...	...	...	...	...	...		
Residual 3/ 9/	**11.5**	**2.7**	**1.1**	**-0.8**	**-0.9**	**-1.2**	**-1.0**	**0.8**	**0.6**	**6.0**	**0.5**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**17.0**	**21.3**	**24.0**	**25.2**	**24.2**	**24.0**	**23.6**	**25.7**	**26.6**		
PV of PPG external debt-to-exports ratio	**68.4**	**75.9**	**79.9**	**80.5**	**76.1**	**75.0**	**73.2**	**79.7**	**82.4**		
PPG debt service-to-exports ratio	**7.4**	**5.6**	**7.4**	**8.4**	**12.3**	**8.4**	**7.0**	**6.8**	**8.1**		
PPG debt service-to-revenue ratio	**7.0**	**5.9**	**8.3**	**9.6**	**14.1**	**9.6**	**7.9**	**7.6**	**9.1**		
Gross external financing need (Million of U.S. dollars)	5,259	8,225	7,654	7,434	8,634	8,818	9,607	14,028	28,958		
Key macroeconomic assumptions											
Real GDP growth (in percent)	1.6	5.0	5.3	5.5	5.5	5.5	5.5	5.1	5.0	**6.1**	**5.3**
GDP deflator in US dollar terms (change in percent)	-1.7	1.2	4.2	5.1	5.6	5.2	5.3	2.5	1.9	**-2.8**	**4.0**
Effective interest rate (percent) 4/	3.1	2.3	3.2	3.4	4.3	3.3	2.7	3.5	3.7	**2.6**	**3.5**
Growth of exports of G&S (US dollar terms, in percent)	-15.4	19.8	17.6	15.4	13.0	11.4	12.1	7.7	7.0	**2.9**	**12.2**
Growth of imports of G&S (US dollar terms, in percent)	-15.2	18.0	13.4	11.6	10.9	10.1	10.4	7.7	7.0	**8.7**	**10.9**
Grant element of new public sector borrowing (in percent)	...	19.8	18.8	18.9	18.9	19.3	19.4	17.3	17.3	**...**	**18.5**
Government revenues (excluding grants, in percent of GDP)	26.6	26.6	27.0	27.3	27.7	28.1	28.6	28.6	28.6	**27.7**	**28.0**
Aid flows (in Million of US dollars) 5/	0.0	960.0	700.0	707.0	707.0	805.0	805.0	548.5	1104.3		
Grant-equivalent financing (in percent of GDP) 6/	...	1.5	1.4	1.2	1.0	0.9	0.8	0.6	0.6	**...**	**0.9**
Grant-equivalent financing (in percent of external financing) 6/	...	19.8	18.8	18.9	18.9	19.3	19.4	17.3	17.3	**...**	**18.5**
Nominal GDP (Million of US dollars)	57,706	61,320	67,283	74,583	83,046	92,155	102,379	157,773	311,343		
Nominal dollar GDP growth	0.0	6.3	9.7	10.8	11.3	11.0	11.1	7.7	7.0	**3.2**	**9.6**
Memorandum items:											
PV of external debt 7/	38.3	42.3	44.7	45.2	43.5	42.4	41.0	43.1	44.0		
In percent of exports	153.6	150.7	148.6	144.3	136.8	132.5	127.1	133.7	136.3		
Total external debt service-to-exports ratio	30.0	38.4	32.5	30.2	34.1	30.4	29.3	29.0	30.3		
PV of PPG external debt (in Million of US dollars)	9,835	13,078	16,179	18,814	20,124	22,086	24,150	40,570	82,677		
(PVt-PVt-1)/GDPt-1 (in percent)		5.6	5.1	3.9	1.8	2.4	2.2	2.1	1.8		
Non-interest current account deficit that stabilizes debt ratio	-10.4	1.3	2.6	5.0	4.7	5.8	6.0	3.5	3.5		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

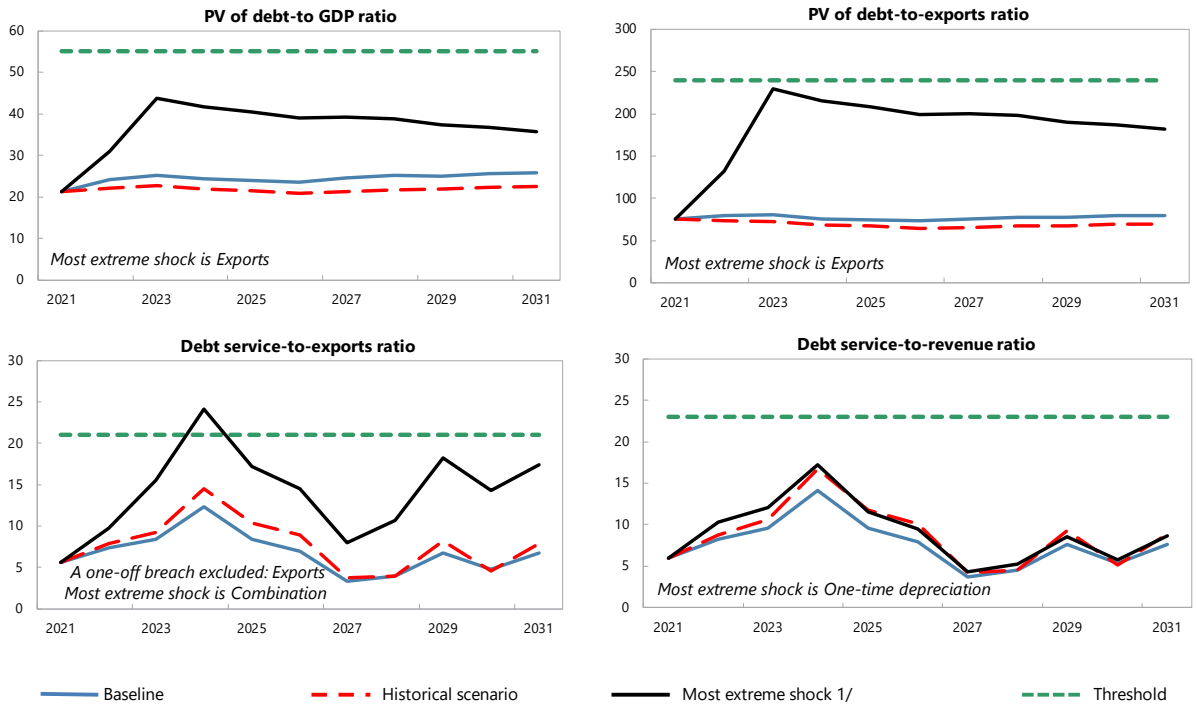
6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

9/ Residual in 2019 is a result of large errors and ommisions, reserve accumulation, and other outflows (mostly households' FX accumulation), while the residual in 2020 is attributed to the use of reserves finance external financing needs.

Figure 1. Uzbekistan: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2021-2031



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices [2/]	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	3.0%	3.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	18	18
Avg. grace period	5	5

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one off breach, only that stress test (with a one-off breach) would be presented.

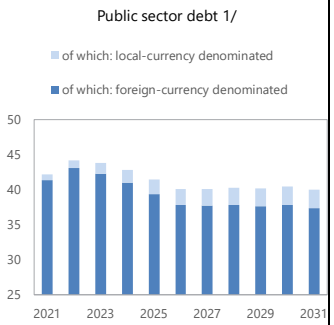
2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

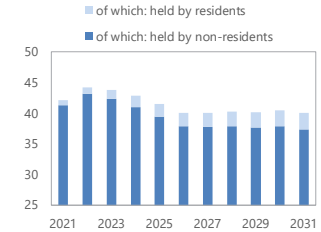
Table 2. Uzbekistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020-2041

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**37.8**	**42.1**	**44.2**	**43.8**	**42.8**	**41.4**	**40.0**	**40.0**	**40.0**	**15.0**	**41.4**
of which: external debt	**37.2**	**41.3**	**43.1**	**42.3**	**41.0**	**39.4**	**37.8**	**37.3**	**34.5**	**15.0**	**39.4**
Change in public sector debt	8.5	4.3	2.1	-0.4	-1.0	-1.4	-1.4	-0.4	-0.2		
Identified debt-creating flows	**8.6**	**3.4**	**1.8**	**-0.6**	**-0.6**	**-1.6**	**-1.8**	**-0.5**	**0.1**	**2.7**	**-0.3**
Primary deficit	**4.2**	**4.2**	**3.9**	**3.4**	**2.1**	**2.3**	**2.0**	**2.0**	**2.0**	**0.0**	**2.6**
Revenue and grants	26.6	26.6	27.0	27.3	27.7	28.1	28.6	28.6	28.6	**27.8**	**28.0**
of which: grants	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Primary (noninterest) expenditure	30.8	30.8	30.9	30.7	29.8	30.5	30.7	30.7	30.7	**27.8**	**30.6**
Automatic debt dynamics	**-0.1**	**-2.2**	**-3.4**	**-4.5**	**-2.8**	**-3.6**	**-3.3**	**-1.9**	**-1.7**		
Contribution from interest rate/growth differential	0.0	-2.3	-2.0	-2.1	-1.5	-2.1	-2.4	-1.7	-1.6		
of which: contribution from average real interest rate	0.4	-0.5	0.1	0.2	0.7	0.1	-0.2	0.2	0.3		
of which: contribution from real GDP growth	-0.5	-1.8	-2.1	-2.3	-2.3	-2.2	-2.2	-2.0	-1.9		
Contribution from real exchange rate depreciation	-0.1	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**4.6**	**1.4**	**1.2**	**0.5**	**0.0**	**-0.3**	**-0.6**	**-0.6**	**-0.2**	**2.6**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (SOEs' Guarantees)	4.6	1.4	1.3	0.5	0.1	-0.3	-0.5	-0.5	-0.2		
Residual	**-0.1**	**0.9**	**0.2**	**0.2**	**-0.3**	**0.2**	**0.4**	**0.1**	**-0.3**	**0.4**	**0.5**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**18.4**	**23.0**	**25.8**	**26.7**	**26.2**	**25.9**	**25.9**	**28.7**	**32.3**		
PV of public debt-to-revenue and grants ratio	**69.1**	**86.5**	**95.7**	**97.8**	**94.4**	**92.2**	**90.3**	**100.4**	**113.0**		
Debt service-to-revenue and grants ratio 3/	**8.7**	**7.9**	**10.5**	**12.0**	**17.7**	**13.9**	**12.7**	**14.3**	**24.8**		
Gross financing need 4/	11.1	7.7	8.0	7.1	7.1	5.9	5.1	5.6	8.9		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	1.6	5.0	5.3	5.5	5.5	5.5	5.5	5.1	5.0	**6.1**	**5.3**
Average nominal interest rate on external debt (in percent)	2.6	0.4	2.0	2.3	3.7	2.2	1.3	2.4	2.7	**2.1**	**2.4**
Average real interest rate on domestic debt (in percent)	28.3	2.0	2.8	3.6	3.2	2.6	2.1	1.9	1.7	**28.3**	**2.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.4	...	...	...	...	...	...	...	...	**9.6**	**...**
Inflation rate (GDP deflator, in percent)	11.9	9.8	11.5	7.9	5.9	5.3	5.3	5.3	5.3	**16.0**	**6.5**
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.7	5.2	5.5	4.8	2.5	7.8	6.2	5.1	5.0	**7.5**	**5.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-4.4	-0.1	1.9	3.8	3.1	3.7	3.4	2.4	2.2	**-2.6**	**2.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The general government, and government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

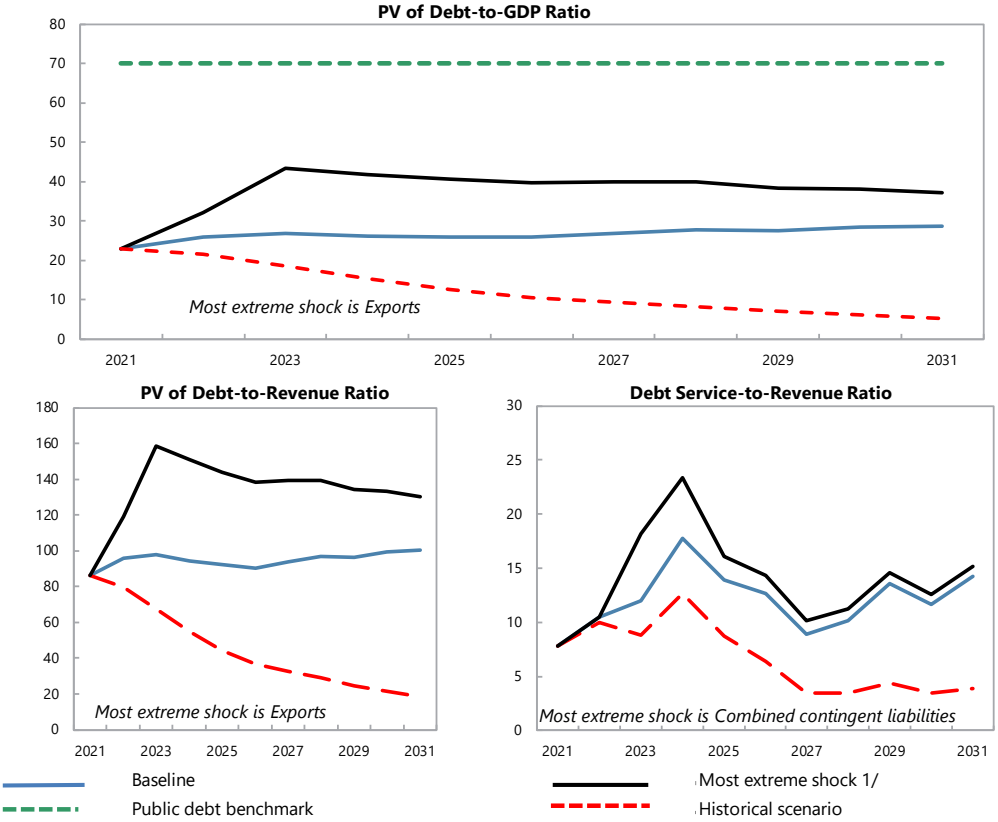
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 2. Uzbekistan: Indicators of Public Debt Under Alternative Scenarios, 2021-2031



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	78%	78%
Domestic medium and long-term	11%	11%
Domestic short-term	16%	11%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	3.0%	3.0%
Avg. maturity (incl. grace period)	18	18
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	2.1%	2.1%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	0.3%	5.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Uzbekistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021-2031

(In percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	21	24	25	24	24	24	24	25	25	26	26
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	21	22	23	22	21	21	21	22	22	22	22
B. Bound Tests											
B1. Real GDP growth	21	25	27	26	25	25	26	27	26	27	27
B2. Primary balance	21	26	31	30	30	29	30	30	30	30	30
B3. Exports	21	31	44	42	40	39	39	39	37	37	36
B4. Other flows 3/	21	32	40	38	37	36	36	36	34	34	33
B5. Depreciation	21	30	28	27	26	26	27	28	29	30	30
B6. Combination of B1-B5	21	36	42	40	39	38	38	38	36	36	35
C. Tailored Tests											
C1. Combined contingent liabilities	21	32	33	33	32	31	32	32	32	32	32
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	21	27	28	28	27	27	28	28	28	28	29
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	76	80	80	76	75	73	76	78	77	79	80
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	76	73	72	69	67	64	66	67	68	69	69
B. Bound Tests											
B1. Real GDP growth	76	80	80	76	75	73	76	78	77	79	80
B2. Primary balance	76	87	100	96	94	91	93	94	92	94	93
B3. Exports	76	133	229	215	208	199	200	198	190	187	182
B4. Other flows 3/	76	105	127	119	115	111	111	111	106	105	103
B5. Depreciation	76	80	70	67	66	65	68	70	71	73	74
B6. Combination of B1-B5	76	140	128	178	172	165	166	164	158	156	152
C. Tailored Tests											
C1. Combined contingent liabilities	76	106	107	103	100	97	99	100	98	99	99
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	76	80	82	78	77	75	77	79	77	79	80
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	6	7	8	12	8	7	3	4	7	5	7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	6	8	9	15	10	9	4	4	8	5	8
B. Bound Tests											
B1. Real GDP growth	6	7	8	12	8	7	3	4	7	5	7
B2. Primary balance	6	7	9	13	9	8	4	5	8	6	8
B3. Exports	6	10	16	**24**	17	15	8	11	18	14	17
B4. Other flows 3/	6	7	9	14	10	8	5	6	10	8	10
B5. Depreciation	6	7	8	12	8	7	3	4	6	4	6
B6. Combination of B1-B5	6	9	14	21	15	12	7	10	15	12	14
C. Tailored Tests											
C1. Combined contingent liabilities	6	7	9	13	9	8	4	5	7	5	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	6	7	9	13	9	7	6	6	9	5	7
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	6	8	10	14	10	8	4	4	8	5	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	6	9	11	17	12	10	4	4	9	5	9
B. Bound Tests											
B1. Real GDP growth	6	8	10	15	10	8	4	5	8	6	8
B2. Primary balance	6	8	10	15	11	9	5	6	9	7	9
B3. Exports	6	8	11	17	12	10	5	7	12	10	12
B4. Other flows 3/	6	8	11	16	11	9	5	7	11	9	11
B5. Depreciation	6	10	12	17	12	9	4	5	9	6	9
B6. Combination of B1-B5	6	9	12	17	12	10	5	8	12	10	12
C. Tailored Tests											
C1. Combined contingent liabilities	6	8	11	15	11	9	5	5	8	6	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	6	8	10	15	10	8	7	7	10	5	8
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

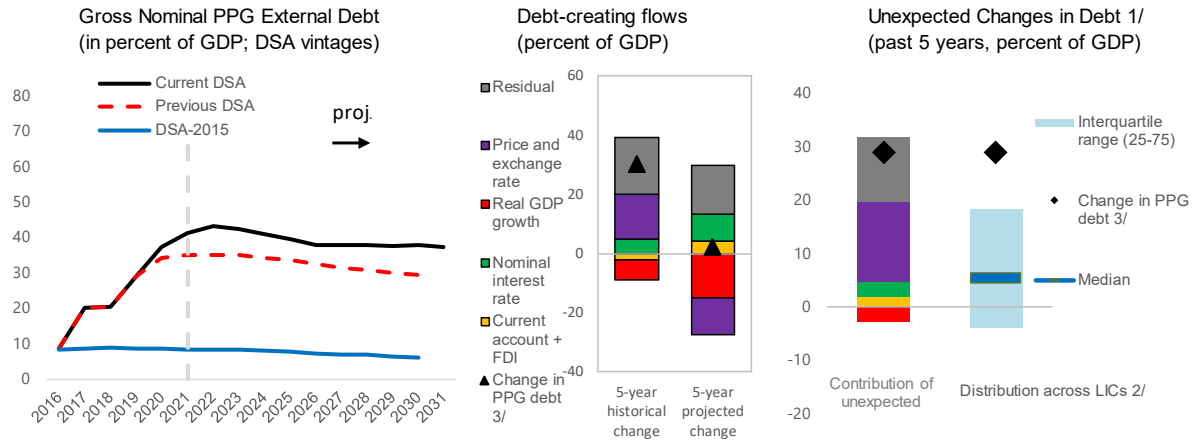
Table 4. Uzbekistan: Sensitivity Analysis for Key Indicators of Public Debt, 2021-2031
(In percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	23	26	27	26	26	26	27	28	28	29	29
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	23	21	19	15	13	10	9	8	7	6	5
B. Bound Tests											
B1. Real GDP growth	23	27	30	31	32	33	35	37	38	40	42
B2. Primary balance	23	29	34	33	32	32	32	33	33	33	33
B3. Exports	23	32	43	42	41	40	40	40	38	38	37
B4. Other flows 3/	23	34	41	40	39	38	38	38	37	37	36
B5. Depreciation	23	29	27	24	22	21	20	19	18	17	16
B6. Combination of B1-B5	23	26	29	29	28	28	29	29	29	29	29
C. Tailored Tests											
C1. Combined contingent liabilities	23	36	36	35	34	34	34	35	34	35	35
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	23	26	27	27	26	26	27	28	28	28	29
Public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	86	96	98	94	92	90	94	97	97	100	100
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	86	80	68	55	45	37	33	29	24	22	18
B. Bound Tests											
B1. Real GDP growth	86	101	111	112	114	116	123	130	134	141	145
B2. Primary balance	86	106	124	119	114	111	113	115	114	116	116
B3. Exports	86	119	158	151	144	138	139	139	134	133	130
B4. Other flows 3/	86	124	152	144	138	133	134	134	130	129	126
B5. Depreciation	86	107	99	88	79	72	70	68	63	61	57
B6. Combination of B1-B5	86	98	107	103	100	97	100	102	101	103	102
C. Tailored Tests											
C1. Combined contingent liabilities	86	134	132	126	121	117	119	121	120	122	122
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	86	96	99	96	94	92	95	98	97	100	100
Debt Service-to-Revenue Ratio											
Baseline	8	10	12	18	14	13	9	10	14	12	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	8	10	9	13	9	6	3	3	4	3	4
B. Bound Tests											
B1. Real GDP growth	8	11	13	20	16	15	12	13	17	16	19
B2. Primary balance	8	10	14	22	17	14	10	12	16	14	16
B3. Exports	8	10	13	20	16	14	10	13	18	16	18
B4. Other flows 3/	8	10	13	19	16	14	10	13	17	15	18
B5. Depreciation	8	11	14	20	15	14	9	10	13	11	13
B6. Combination of B1-B5	8	10	14	20	16	15	11	12	15	13	15
C. Tailored Tests											
C1. Combined contingent liabilities	8	10	18	23	16	14	10	11	15	13	15
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	8	10	12	18	14	13	12	13	16	12	14

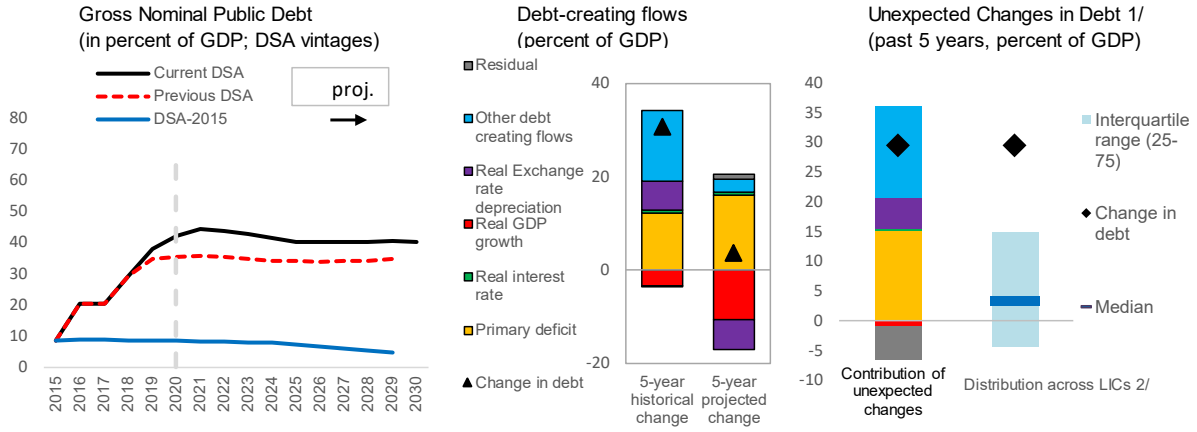
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Uzbekistan: Drivers of Debt Dynamics—Baseline Scenario

External debt



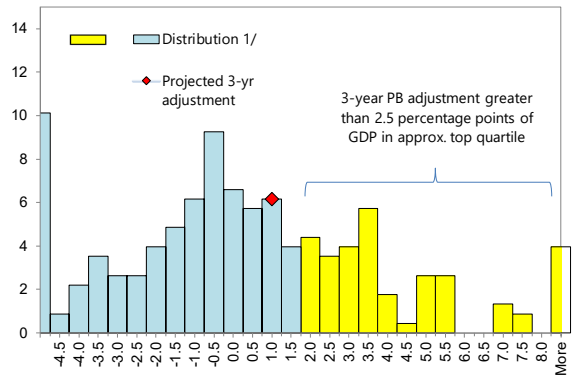
Public debt



1/ Difference between anticipated and actual contributions on debt ratios.
2/ Distribution across LICs for which LIC DSAs were produced.
3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.
Source: IMF staff estimates.

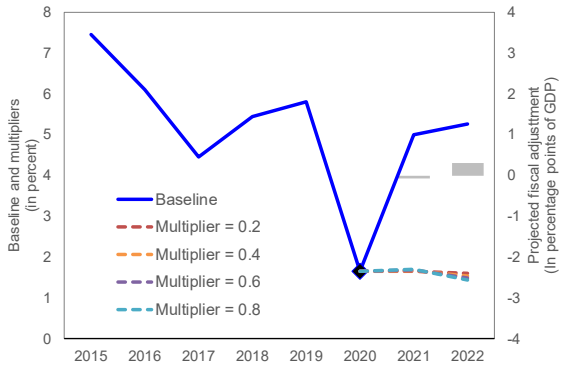
Figure 4. Uzbekistan: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



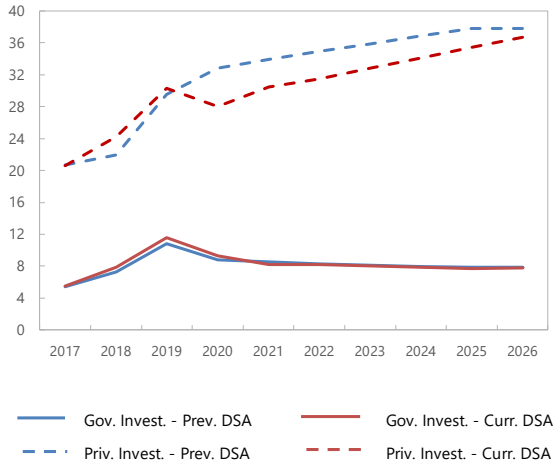
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

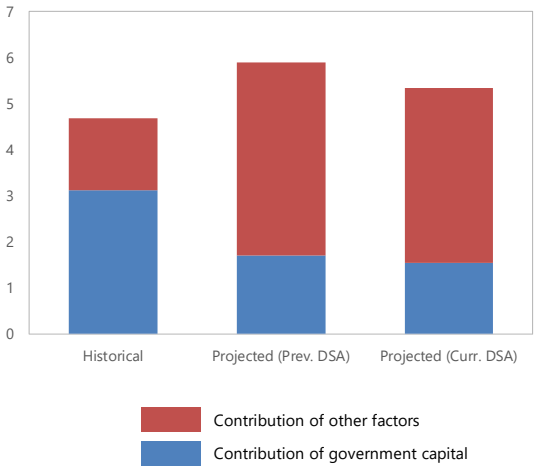
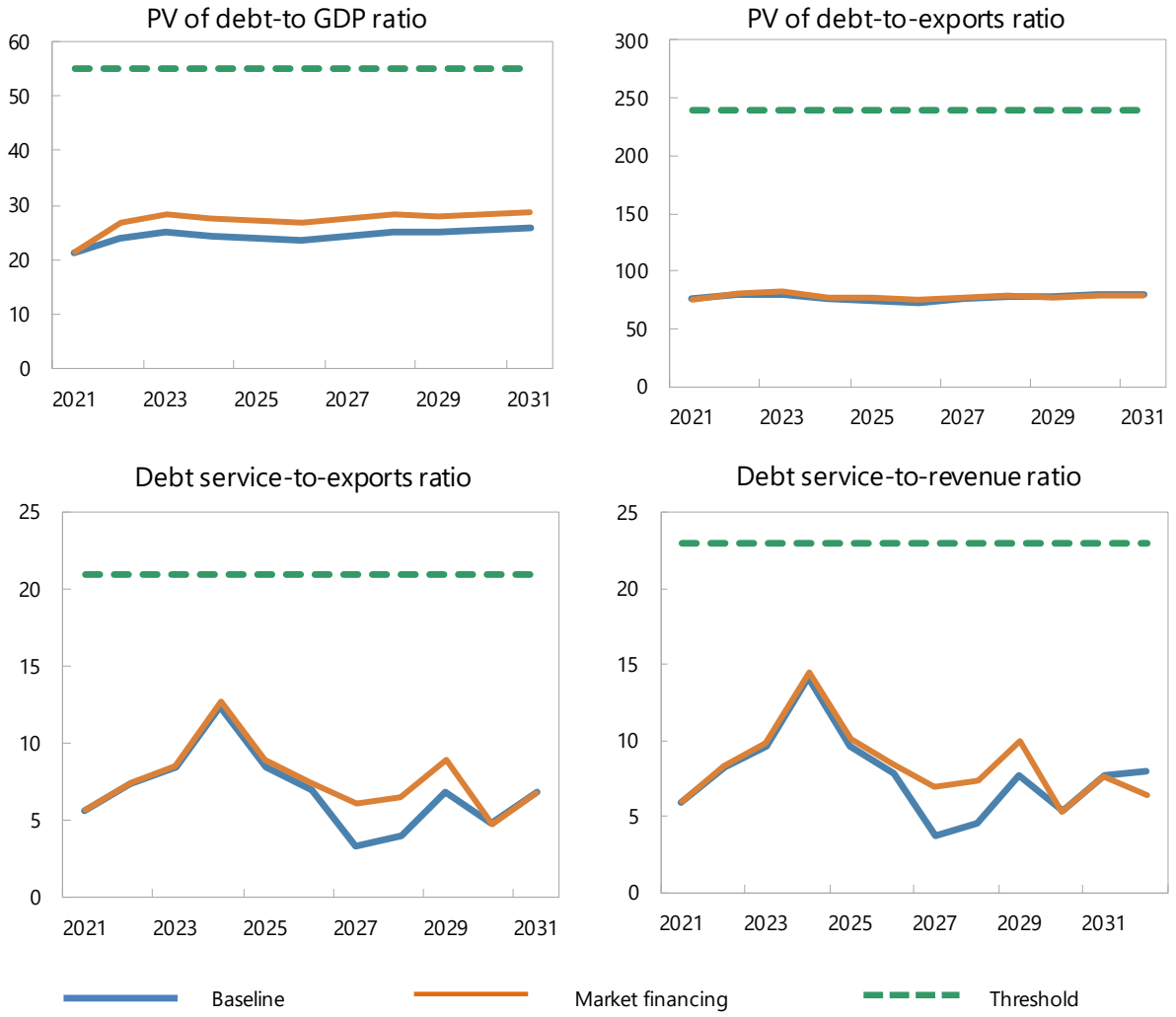


Figure 5. Uzbekistan: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	8	267
Breach of benchmark	No	No
Potential heightened liquidity needs	Low	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.

2/ Uzbeksitan is not included in EMBI. Spread from the Uzbek 2019 international issuance.



Sources: Country authorities; and staff estimates and projections.